# Multibranch Feature Fusion Network With Self- and Cross-Guided Attention for Hyperspectral and LiDAR Classification

Wenqian Dong, *Member, IEEE*, Tian Zhang, Jiahui Qu, *Member, IEEE*, Song Xiao, *Member, IEEE*, Tongzhen Zhang, and Yunsong Li, *Member, IEEE*

***Abstract*— The effective fusion of multisource data helps to improve the performance of land cover classification. Most existing convolutional neural network (CNN)-based methods adopt an early/late fusion strategy to fuse the low-/high-level features for classification, which still has two inherent challenges: 1) the conventional convolution operation performs a weighted average operation on each pixel in the receptive field, which will reduce the discriminability of the center pixel due to the influence of the interference pixels and 2) the spatial–spectral features of the hyperspectral image (HSI), the elevation features of light detection and ranging (LiDAR), and the complementary features between the multimodal data are not fully exploited, which results in the reduction of classification accuracy. In this article, an effective multibranch feature fusion network with self- and cross-guided attention (MB2FscgaNet) is proposed for the joint classification of LiDAR and HSI. The main concern of this article is how to accurately estimate more effective spectral–spatial-elevation features and yield more effective transfer in the network. Specifically, MB2FscgaNet adopts a multibranch feature fusion architecture to fully exploit the hierarchical features from LiDAR and HSI level by level. At each level of the network, a self- and cross-guided attention (SCGA) is developed to assign a higher weight to interesting areas and channels of LiDAR and HSI feature maps to obtain refined spectral–spatial-elevation features and provide complementary information cross-guidance between LiDAR and HS. We further designed a spectral supplement module (SeSuM) to improve the discriminative ability of the center pixel. Comparative classification results and ablation studies demonstrate that the proposed MB2FscgaNet achieves competitive performance against state-of-the-art methods.**

***Index Terms*— Hyperspectral image (HSI), joint classification, light detection and ranging (LiDAR), self- and cross-guided attention (SCGA).**

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Wenqian Dong, Tian Zhang, Jiahui Qu, Tongzhen Zhang, and Yunsong Li are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: wqdong@xidian.edu.cn; 20011210545@stu.xidian.edu.cn; jhqu@xidian.edu.cn; tzzhang_1@stu.xidian.edu.cn; ysli@mail.xidian.edu.cn).

Song Xiao is with the Beijing Electronic Science and Technology Institute, Beijing 100070, China, and also with the School of Communication Engineering, Xidian University, Xi'an 710071, China (e-mail: xiaosong@mail.xidian.edu.cn).

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## I. Introduction

MODERN advanced sensor technology has made it possible to acquire different remote sensing images, such as hyperspectral image (HSI), light detection and ranging (LiDAR), and synthetic aperture radar (SAR), from different sensors simultaneously [1], [2]. Different remote sensing images obtained from the same geographical area can provide different information on ground features and are widely used in environmental monitoring [3], urban-growth analysis [4], and agricultural monitoring [5], [6]. Among these multisource remote sensing data, HSI with abundant spectral information has a powerful ability to distinguish different materials [7], [8], which has been widely used in obtaining satisfactory detection [9] and classification results [10], [11]. However, objects of different altitudes with the same material cannot be classified by HSI due to similar spectral responses [12], [13]. Different from HSI data, LiDAR-DSM provides elevation information for scenes [14], thus making it possible to accurately distinguish the classes within different heights. Based on the above facts, it is possible to obtain higher classification accuracy by fusing HSI and LiDAR data. Fig. 1 illustrates an application example of our proposed multibranch feature fusion network with self- and cross-guided attention (MB2FscgaNet) in the joint classification of HSI and LiDAR. Experiments show that classification on the fused HSI can, indeed, improve classification accuracy.

In recent years, a large number of classification strategies have been developed to fuse HSI and LiDAR images, and they have been applied for land-cover and land-use classification [15], such as forest preservation and management [16], and microclimate modeling [17]. One of the most widely used methods is feature-level-based. Morphological extended attribute profiles (EAPs) were put into use in [18] to stack the HSI and LiDAR data together for classification. Nevertheless, the direct concatenation of the high-dimensional features may cause the curse of dimensionality and then lead to

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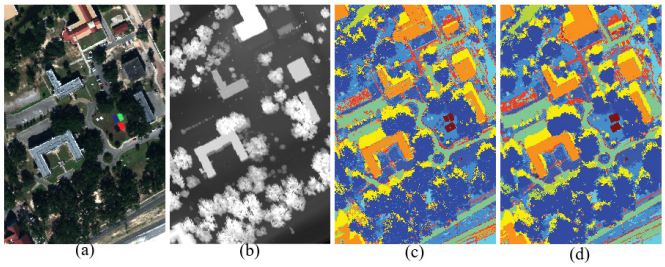



Fig. 1. Application example of our proposed MB2FscgaNet for the MUUFL dataset. (a) Pseudocolor image for HSI. (b) Gray image for LiDAR. (c) Classification map by HSI. (d) Classification map by HSI + LiDAR.

the Hughes phenomenon [19]. The problem occurs especially when the number of available training data is small. The principal component analysis (PCA) was employed to address the aforementioned problem [20]. PCA is able to project the multisource data into the common subspace and exploit the variance in the principal components, thus reducing the dimensionality. Hong *et al.* [21] proposed a manifold learning strategy for dimensionality reduction, which is based on local linear embedding, and achieved a favorable performance for HSI classification. Rasti *et al.* [22] used orthogonal total variation component analysis (OTVCA) to estimate the high dimension features in a lower dimensional space and produce accurate classification maps. Another popularly adopted fusion strategy is based on the decision level. With nonlinear kernel functions, support vector machines (SVMs) [23] have become an effective tool for classification. Taking [24] as an example, four classifiers were generated by the SVM with the elevation features, spatial features, spectral features, and the fusion of them as the inputs. Besides the SVMs, the classification method using random forest (RF) is another novel ensemble way. A majority voting method was introduced in RF and is advantageous in classifying due to its tolerance of big data [25]. All the above methods have shown satisfying classification performance. However, these classical methods depend heavily on the parameter setting, and it is time-consuming to select an optimal parameter for the different classification tasks whether based on feature level or decision level.

In the past few years, classification technologies of remote sensing data based on deep learning have attracted wide attention [26], [27]. Among deep learning strategies, a convolutional neural network (CNN), which is especially suitable for solving HSI and LiDAR classification tasks for its strong learning ability, has become an effective tool for classification. Most of the existing CNN-based methods employ a two-branch network to extract the spatial–spectral information of HSI and the elevation information of LiDAR, respectively. Specifically, they can be divided into two continuous processes: 1) extract features from HSI images and LiDAR images by a two-branch network and 2) input the extracted features into a classifier to generate the classification result. Such a two-branch network facilitates feature extraction, which yielded great improvement in the accuracy of classification tasks. Some researchers have employed CNN to classify and effectively improve the performance. CNN-based classification approaches are highly capable of extracting robust high-level features [28]. Chen *et al.* [29] introduced a deep feature extraction approach, which used CNN for HSI classification by adopting numerous pooling and convolutional layers to extract deep features in HSI. In order to improve the classification performance of HSI, a spectral–spatial capsule network (CapsNet) architecture based on CNN was introduced in [30], which significantly reduces the complexity of the network. Chen *et al.* [31] designed an end-to-end CNN model to extract deep features from HSI and LiDAR data, then utilized a fully connected CNN to fuse the extracted features, and, finally, adopted logistic regression to generate the final classification results. Xu *et al.* [17] also introduced a spectral–spatial CNN-based two-branch framework, which is equipped with cascaded blocks for extracting features from HSI and LiDAR images. Chen *et al.* [32] merged the LiDAR and HSI data and designed a dual-tunnel CNN framework to perform classification on the merged data.

Considering that interfering pixels with different classification labels from the center pixel in the convolution operation will reduce the distinguishability of the center pixel, some researchers apply the attention mechanism to the classification task to suppress the influence of the interfering pixels. The attention mechanism can help the model assign different weights to each part of the input and focus on the area of interest. It ulteriorly helps the model make more accurate judgments without incurring greater costs for the calculation and storage of the model. A typical attention model on a single image is squeeze-and-excitation network (SENET) [33]. As the world champion of ImageNet 2017, SENET can be embedded into almost any popular network. Ashish *et al.* [34] developed a transformer to draw global dependencies between input and output. What is more, a spectral–spatial attention network (SSAN) for HSI classification was proposed in [35], which contains an attention module for exploring effective features of HSI cubes, effectively improving the classification accuracy.

The HSI with abundant spectral information and the LiDAR data that can provide scene elevation information are multimodal and heterogeneous data, and have different data characteristics. Due to the different characteristics of HSI and LiDAR, how to fully extract the multimodal features of HSI and LiDAR for accurate classification is a problem that needs to be deeply explored. The CNN-based methods can extract multimodal features from the HS and LiDAR data. However, many CNN-based HSI and LiDAR classification methods do not fully exploit the spatial–spectral information of HSI, the elevation information of LiDAR, and the complementary guidance information between the multimodal data, which leads to the reduction of classification accuracy. Some CNN-based HSI and LiDAR classification networks design each stream to process a single modal image and simply connect multiple modes to a stream [17], [31], [36]. The work in [32] directly fused the LiDAR and HSI data and used the fused image as the input of the CNN network to extract features. These existing CNN-based methods mainly focus on early/late fusion strategy, which is difficult to discover the complementary guidance

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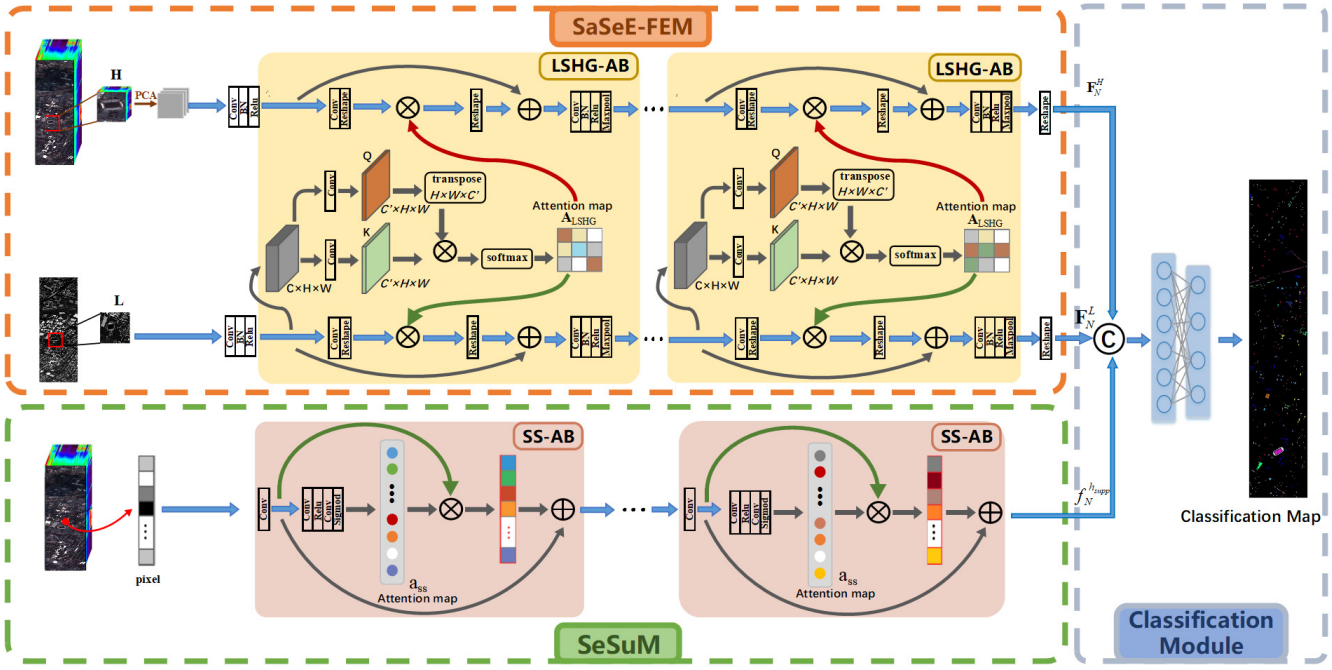



Fig. 2. Flowchart of the proposed MB2FscgaNet. As shown in the figure, multisource images are fed into the spatial–spectral-elevation feature extraction module (SaSeE-FEM) and the spectral supplement module (SeSuM) to obtain refined spatial–spectral-elevation-enhanced features. The LiDAR-self- and HSI-guided-attention block (LSHG-AB) uses the LiDAR-derived attention mask to provide guidance for spectral-spatial-elevation information extraction. The self-spectral-AB (SS-AB) builds the relationships between channels to calibrate channelwise features. The outputs are then cascaded and sent to the classification module for pixel classification.

relationship between HSI and LiDAR. So far, the impact of guidance across multimodal images, and its application in the joint classification of HSI and LiDAR, remains unexplored.

In this article, an MB2FscgaNet is proposed for the classification of multimodal remote sensing data, which has more compact complementary information blending and more effective information transfer for the multimodal features of HSI and LiDAR data, resulting in better classification performance. The major contributions of this article are summarized as follows.

1) An MB2FscgaNet is proposed for joint classification of multisource data, which has total freedom to learn complementary spatial–spectral-elevation information between the modalities. In the proposed MB2FscgaNet, LSHG-ABs are developed to be embedded in the SaSeE-FEM to efficiently transfer and mine the complementary spatial–spectral-elevation features between the two streams of HSI and LiDAR. An SeSuM is added to the proposed network as a spectral supplementary branch to further improve the distinguishability of the center pixel.
2) An effective self- and cross-guided attention (SCGA) is proposed to be embedded in the proposed MB2FscgaNet network. SCGA not only refines the multimodal features extracted from HSI and LiDAR streams but also provides complementary information cross-guidance from the LiDAR stream to the HSI stream, which exploits the advantages of one multimodal data to complement the weaknesses of another and yields more effective spectral–spatial-elevation information blending in the network.

This article is organized as follows. Section II presents the details of the proposed MB2FscgaNet. The descriptions of datasets and experimental results are given in Section II. This article is finally concluded in Section V.

## II. Methodology

### A. Architecture Overview

The goal of the classification of HSI and LiDAR is to make full use of the joint information of these two data sources for fine classification of the Earth's surface. Just as with any other classification tasks, how to fully extract and fuse complementary information that is beneficial to classification tasks of HSI and LiDAR data without being affected by the Hughes effect is the key to improving classification accuracy. In this article, we design an MB2FscgaNet for better classification performance by jointly extracting the features of HSI and LiDAR data.

The overall architecture of the proposed MB2FscgaNet is represented in Fig. 2. HSI has a powerful ability to distinguish multiple materials based on their rich spectral features. Different from HSI data, LiDAR data can describe the height and shape of the targets through the rich elevation information contained in it. Sufficiently fusing these complementary features is a critical step for HSI and LiDAR data collaborative classification. An SaSeE-FEM (marked with an orange box in Fig. 2) is designed for extracting the complementary spatial–spectral-elevation features from multidata source HSI and LiDAR. LSHG-ABs are further designed to obtain more effective spectral–spatial-elevation features of all layers and yield more effective transfer in the network. In addition,

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an SeSuM (marked with the green box in Fig. 2) is designed for improving the discriminative ability of the center pixel, in which SS-ABs are designed for learning the interchannel relationships of HSI.

### *B. Spatial–Spectral-Elevation Feature Extraction Module*

It is well-known that HSI data-carrying plenty of spectral information could present multiple spectral characteristics of land covers, while LiDAR images consist of both reflectance properties and 3-D geospatial information of the surveyed area. This motivates us to build a feature fusion module to fully exploit the complementary information from two data sources. An SaSeE-FEM (marked with the orange box in Fig. 2) is designed to fully capture the hierarchical spatial–spectral and elevation features from HSI and LiDAR features. More precisely, SaSeE-FEM is a cascade of $N$ LSHG-ABs, which provides significant spectral–spatial-elevation information guidance within and from LiDAR image to HSI and, thus, obtains refined spatial–spectral-elevation-enhanced features, thereby yielding more effective transfer in the network.

Given the HSI patch $\mathbf{H} \in \mathbb{R}^{H\times W\times C}$ and the LiDAR patch $\mathbf{L} \in \mathbb{R}^{H\times W\times 1}$ that cover the same area as the inputs of the SaSeE-FEM, where $H$, $W$, and $C$ represent the height, width, and the number of spectral bands of the patch, respectively, the PCA is first used on $\mathbf{H}$ for reducing the redundant spectral information. Furthermore, $N$ LSHG-AB is cascaded for effectively integrating multilevel features. Let $\mathbf{F}_N^H$ and $\mathbf{F}_N^L$ be the output feature maps of the SaSeE-FEM; they can be formulated by

$$\begin{cases} \mathbf{F}_N^H = f_{\text{LSHG-AB},N}, \ldots, f_{\text{LSHG-AB},i}, \cdots \\ \qquad f_{\text{LSHG-AB},1}[f_{\text{conv}}(\mathbf{H}_{\text{PCA}})] \\ \mathbf{F}_N^L = f_{\text{LSHG-AB},N}, \ldots, f_{\text{LSHG-AB},i}, \cdots \\ \qquad f_{\text{LSHG-AB},1}[f_{\text{conv}}(\mathbf{L})] \end{cases} \tag{1}$$

where $\mathbf{H}_{\text{PCA}}$ represents the HSI patch after PCA, $f_{\text{LSHG-AB},i}$ denotes the function of the $i$th LSHG-AB, and $f_{\text{conv}}$ represents the convolution operation.

As shown in the yellow area in Fig. 2, the LiDAR-derived attention mask of LSHG-AB captures nonlocal contextual relations between each pixel and all other pixels. LSHG-AB further uses the LiDAR-derived attention mask to provide guidance for spectral–spatial-elevation information extraction. The attention mask is utilized not only to add relevant elevation information to the LiDAR stream but also to enhance the spatial features that went "unnoticed" in HSI. Such an attention mechanism provides the refined feature maps containing more discriminative spatial–spectral-elevation features, enabling the network to perform more effective transmission.

For the $i$th LSHG-AB, the inputs are expressed as $\mathbf{F}_{i-1}^H$ and $\mathbf{F}_{i-1}^L$, which are also the outputs of the previous module. $\mathbf{F}_{i-1}^H$ and $\mathbf{F}_{i-1}^L$ are first passed through a convolution operation and reshaped into 2-D weighted feature maps. The obtained feature maps are then multiplied with attention map $\mathbf{A}_{\text{LSHG}}$ and added to the previous unemphasized input $\mathbf{F}_{i-1}^H$ and $\mathbf{F}_{i-1}^L$ so that the spatial–spectral-elevation information is sufficiently enhanced, and thus, the spatial–spectral-enhanced feature map of HSI and the spatial-enhanced feature map of LiDAR are obtained. Then, both two feature maps are fed into a convolutional operator to further obtain a discriminative representation. This convolutional layer is followed by a batch normalization (BN) layer to regularize the training process, and then, a rectified liner unit (ReLu) layer is followed to learn a nonlinear representation. Finally, in order to reduce the computation complexity and the data variance, the feature map is sent to a maxpooling (MP) layer. Mathematically, the outputs of the $i$th LSHG-AB can be formulated by

$$\begin{cases} \mathbf{F}_i^H = f_{\text{LSHG-AB},i}\left(\mathbf{F}_{i-1}^H\right) \\ \qquad = f_{\text{conv}}\left[\left(f_{\text{conv}}\left(\mathbf{F}_{i-1}^H\right) \otimes \mathbf{A}_{\text{LSHG},i}\right) \oplus \mathbf{F}_{i-1}^H\right] \\ \mathbf{F}_i^L = f_{\text{LSHG-AB},i}\left(\mathbf{F}_{i-1}^L\right) \\ \qquad = f_{\text{conv}}\left[\left(f_{\text{conv}}\left(\mathbf{F}_{i-1}^L\right) \otimes \mathbf{A}_{\text{LSHG},i}\right) \oplus \mathbf{F}_{i-1}^L\right] \\ \qquad\qquad (i = 1, 2, \ldots, N) \end{cases} \tag{2}$$

where $\otimes$ and $\oplus$ represent the elementwise multiplication operation and the broadcasting addition operation, respectively, and $\mathbf{A}_{\text{LSHG},i}$ represents the attention map, which is calculated from $\mathbf{F}_{i-1}^L$. Specifically, $\mathbf{F}_{i-1}^L$ is entered into two branches separately and performed two convolution operation to obtain features $\mathbf{Q}$ and $\mathbf{K}$. Furthermore, $\mathbf{Q}$ is transposed and performed the matrix multiplication operation with $\mathbf{K}$. The obtained matrix $\mathbf{Q}^{\text{T}}\mathbf{K}$ with size $HW \times HW$ measures the similarity between each pixel and other surrounding pixels. $\mathbf{Q}^{\text{T}}\mathbf{K}$ is then subjected to softmax operation to obtain the final attention map $\mathbf{A}_{\text{LSHG},i}$. The above process can be more intuitively formulated as

$$\mathbf{A}_{\text{LSHG},i} = \text{softmax}(\mathbf{Q}^{\text{T}}\mathbf{K}) \tag{3}$$

in which

$$\begin{aligned} \mathbf{Q} &= f_{\text{conv}}\left(\mathbf{F}_{i-1}^L\right) \\ \mathbf{K} &= f_{\text{conv}}\left(\mathbf{F}_{i-1}^L\right) \end{aligned} \tag{4}$$

where $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{H\times W\times C'}$ represent local features of the input patch and softmax$(\cdot)$ means the softmax function that is used to normalize the attention weight.

### *C. Spectral Supplement Module*

HSI with rich spectral information and more imaging channels has a powerful ability to distinguish objects. The performance of conventional CNN classification methods is always affected by interfering pixels. In order to eliminate the influence of interference pixels and further improve the discriminative ability of the center pixel, we design an SeSuM to extract spectral information from HSI data as a supplement, which will create spectral-enhanced feature representation of the center pixel. SeSuM is a cascade of $N$ SS-ABs that builds the relationships between channels to calibrate channelwise features. Given the pixel $\boldsymbol{h}_{\text{supp}} \in \mathbb{R}^{1\times C}$ to be classified as the input of SeSuM, the output $\boldsymbol{f}_N^{h_{\text{supp}}}$ of SeSuM can be formulated by

$$\boldsymbol{f}_N^{h_{\text{supp}}} = f_{\text{SS-AB},N}, \ldots, f_{\text{SS-AB},i}, \ldots, f_{\text{SS-AB},1}[f_{\text{conv}}(\boldsymbol{h}_{\text{supp}})] \tag{5}$$

where $\boldsymbol{h}_{\text{supp}}$ is the input vector and $f_{\text{SS-AB},i}$ denotes the function of the $i$th SS-AB. The structure of the proposed

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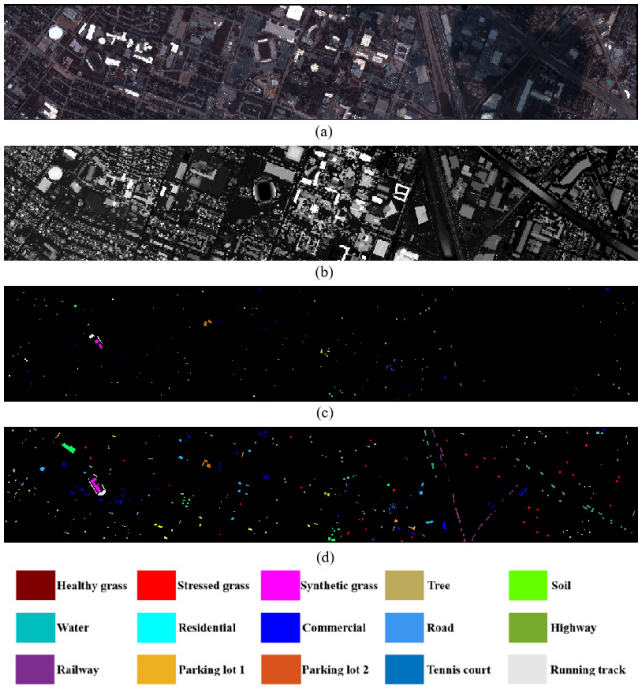



Fig. 3. Visualization of the Houston dataset. (a) Pseudocolor image for the HSI. (b) Gray-scale image for the LiDAR. (c) Training data. (d) Testing data.

SS-AB is shown in Fig. 2 and marked with pink area. SS-AB draws its attention mask from the HSI and can further improve the representative ability of the stream by highlighting the channel of interest. The input of SS-AB is multiplied with the attention map $\boldsymbol{a}_{\mathrm{SS},i}$ to get the highlighted spectral feature. Let $\boldsymbol{f}_{i-1}^{h_{\mathrm{supp}}}$ denote the input of the $i$th SS-AB, and the forward propagation of SS-AB that is element-based can be represented as

$$\begin{aligned}\boldsymbol{f}_{i}^{h_{\mathrm{supp}}} &= f_{\mathrm{SS-AB},i}\left(\boldsymbol{f}_{i-1}^{h_{\mathrm{supp}}}\right)\\ &= \left[\boldsymbol{a}_{\mathrm{SS},i} \otimes f_{\mathrm{conv}}\left(\boldsymbol{f}_{i-1}^{h_{\mathrm{supp}}}\right)\right] \oplus f_{\mathrm{conv}}\left(\boldsymbol{f}_{i-1}^{h_{\mathrm{supp}}}\right)\\ &\qquad (i = 1, 2, \ldots, N) \end{aligned} \tag{6}$$

where $\boldsymbol{f}_{i}^{h_{\mathrm{supp}}}$ denotes the output of the $i$th SS-AB and $\boldsymbol{a}_{\mathrm{SS},i}$ represents the attention map, whose solving process can be expressed as

$$\boldsymbol{a}_{\mathrm{SS},i} = f_{\mathrm{conv}}\left(f_{\mathrm{conv}}\left(\boldsymbol{f}_{i-1}^{h_{\mathrm{supp}}}\right)\right). \tag{7}$$

### D. Classification Module

At the classification stage, the outputs of SaSeE-FEM and SeSuM are concatenated together to further integrate the complementary features, and then, the combined features are fed to a fully connected layer for obtaining the classification map. The classification map can be computed using

$$\mathbf{Map} = \mathrm{fc}\left(\mathbf{F}_N^H L \,\|\, \mathbf{F}_N^L \,\|\, \boldsymbol{f}_N^{h_{\mathrm{supp}}}\right) \tag{8}$$

where $\|$ represents the concatenation operation and fc means the fully connected operation. Eventually, the classification label of the testing image can be denoted as

$$\mathrm{class}(\mathbf{Map}) = \underset{k}{\mathrm{argmax}} \ \|\mathbf{Y}_k\| \tag{9}$$

where $\mathbf{Y}_k$ is the label matrix, in which each element represents the predicated land cover type of the corresponding central pixel of the patch.

TABLE I

NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON DATASET

| Class | | Number of Samples | |
|---|---|---|---|
| | | Train | Test |
| 1 | Health grass | 198 | 1053 |
| 2 | Stressed grass | 190 | 1064 |
| 3 | Synthetic grass | 192 | 505 |
| 4 | Tress | 188 | 1056 |
| 5 | Soil | 186 | 1056 |
| 6 | Water | 182 | 143 |
| 7 | Residual | 196 | 1072 |
| 8 | Commercial | 191 | 1053 |
| 9 | Road | 193 | 1059 |
| 10 | Highway | 191 | 1036 |
| 11 | Railway | 181 | 1054 |
| 12 | Parking lot 1 | 192 | 1041 |
| 13 | Parking lot 2 | 184 | 285 |
| 14 | Tennis court | 181 | 247 |
| 15 | Running track | 187 | 473 |
| | Total | 2832 | 12197 |

### E. Loss Function

The proposed structure is trained by the given training set $\{(x_i, y_i)|i = 1, 2, \ldots, N\}$, where $x_i$ refers to the input $i$th training sample, $y_i$ represents the ground truth of the corresponding $i$th sample, and $N$ is the total number of the training samples. We utilize the cross-entropy loss function to compute the loss values. The possibilities of classes label of each input training patch are defined as

$$p_n = \frac{\exp(\hat{p}_n)}{\sum_k \exp(\hat{p}_k)} \tag{10}$$

where $\hat{p}_n$ represents the nonnormalized output score of the previous layer in the network. In addition, a predicted label will be calculated by the loss function $L$ as

$$L = -\sum_n y_n \log(p_n). \tag{11}$$

## III. EXPERIMENT RESULTS AND ANALYSIS

### A. Datasets' Description

The Houston dataset and the Trento dataset are considered in this article to evaluate the performance of the proposed MB2FscgaNet. The numbers of training and testing samples in each class for these two datasets are demonstrated in Tables I and II.

*1) Houston Dataset*[1]*:* As a typical HSI classification dataset, the Houston dataset was obtained by CASI-1500 on June 23, 2012, covering the University of Houston campus and its surrounding area. This dataset consists of HSI

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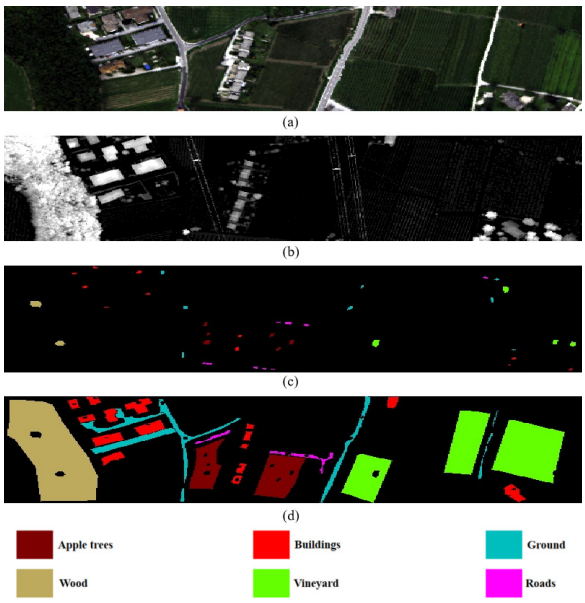



Fig. 4. Visualization of the Trento dataset. (a) Pseudocolor image for the HSI. (b) Gray-scale image for the LiDAR. (c) Training data. (d) Testing data.

images and LiDAR-based DSM. Both the HSI images and LiDAR-based DSM are spatially resampled to 2.5 m with a size of 349 × 1905 pixels. For this image group, the HSI contains 144 spectral bands with a wavelength range of 0.38–1.05 $\mu$m. Fig. 3 shows the visualization of the Houston data. Based on the diverse characteristics, the Houston dataset is divided into 16 categories. The available training samples are divided into 15 classes. On the Houston dataset, 18% of the labeled pixels are selected for training and the remaining for testing. The numbers of training and testing samples are shown in Table I.

*2) Trento Dataset:* The Trento dataset was collected in a rural area in the southern part of Trento, Italy [37]. The HSI and LiDAR-based DSM data are collected by the AISA Eagle sensor and the Optech ALTM 3100EA sensor, respectively. The spatial resolution of these two data is 1 m, and the size is 600 × 166 pixels. The HSI is composed of 63 bands, ranging from 0.42 to 0.99 $\mu$m. Fig. 4 visualizes this data. There exist six different classes. On the Trento dataset, 2% of the labeled pixels are chosen as training samples and the rest as testing samples. The number of training samples and testing samples in each class is shown in Table II.

### B. Experiment Setup

To verify the validity and accuracy of the proposed model, we comprehensively made a comparison between the proposed model, and some classical and state-of-the-art classification methods. The average accuracy (AA), the overall accuracy (OA), and the kappa coefficient (kappa) are applied to evaluate the classification performance of all the models. AA defines the average accuracy for all categories. OA is the ratio of the number of correctly classified pixels to the number of all pixels, calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \times 100\% \tag{12}$$

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE TRENTO DATASET

| Class | | Number of Samples | |
|---|---|---|---|
| | | Train | Test |
| 1 | Apple | 129 | 3905 |
| 2 | Buildings | 125 | 2778 |
| 3 | Ground | 105 | 374 |
| 4 | Woods | 154 | 8969 |
| 5 | Vineyard | 184 | 10317 |
| 6 | Roads | 122 | 3525 |
| | Total | 819 | 29595 |

where TP denotes the true positives, TN denotes the true negatives, the false positives are expressed by FP, and the false negatives are represented by FN. kappa is used as an evaluation index to compare the observed accuracy with the expected accuracy, which is defined as follows:

$$\mathrm{kappa} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \times 100\% \tag{13}$$

where

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} + \frac{(\mathrm{FN} + \mathrm{TN})(\mathrm{FP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}. \tag{14}$$

For the kappa coefficient, as outlined in reference, $<0.4$ represents poor accuracy, 0.4–0.6 signifies moderate accuracy, 0.6–0.8 expresses good accuracy, and $>0.8$ means that the ground truth and the classification result reach a strong agreement.

The deep learning programs are accomplished using the PyTorch framework. All of the deep learning models are implemented on a computer equipped with Intel[1] Core[2] CPU i9-10900x, 64-GB RAM, and NVIDIA RTX3090. The Adam algorithm is used to optimize all of the deep learning models. In the experiments, the learning rate, the batch size, and the patch size are set as 0.00001, 8, and 11 × 11, respectively. The number of training epochs is set as 100, and the numbers of LSHG-AB and SS-AB are set to 3. Several experiments are used to prove the rationality of the parameter settings in the proposed MB2FscgaNet.

### C. Classification Performance

In order to verify the effectiveness of the proposed MB2FscgaNet for the collaborative classification of multisource remote sensing data, we compared the proposed method with several classical and state-of-the-art classifiers, such as SVM, extreme learning machines (ELMs), a deep learning-based two-branch CNN [17], and the recently proposed encoder–decoder network (Endnet) [38]. For a fair comparison, all the competing methods are implemented with optimal parameters. As for the two-branch CNN method, the batch size is set to 100, the number of training epochs is

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TABLE III
CLASS SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE HOUSTON DATASET

| | SVM (H) | SVM (H+L) | ELM (H) | ELM (H+L) | Two-Branch CNN (H) | Two-Branch CNN (H+L) | Endnet (H+L) | Proposed (H+L) |
|---|---|---|---|---|---|---|---|---|
| Health grass | 84.89 | 85.13 | 85.29 | 85.77 | 85.22 | 85.70 | 81.29 | 82.15 |
| Stressed grass | 83.97 | 83.73 | 91.15 | 86.44 | 87.33 | 77.45 | 84.21 | 83.65 |
| Synthetic grass | 100 | 99.86 | 100 | 100 | 99.71 | 99.86 | 100 | 99.80 |
| Tress | 91 | 91.4 | 92.85 | 93.41 | 86.99 | 93.98 | 91.29 | 94.41 |
| Soil | 93.88 | 96.46 | 99.52 | 99.52 | 96.62 | 99.92 | 100 | 100 |
| Water | 97.85 | 97.85 | 97.85 | 97.85 | 92.64 | 92.02 | 93.71 | 98.60 |
| Residual | 86.83 | 89.91 | 91.25 | 83.91 | 81.25 | 87.16 | 86.01 | 90.21 |
| Commercial | 53.78 | 56.67 | 62.22 | 69.53 | 71.24 | 85.30 | 82.05 | 93.54 |
| Road | 78.12 | 67.65 | 77.96 | 80.51 | 71.67 | 86.67 | 86.78 | 86.31 |
| Highway | 60.72 | 63.98 | 62.51 | 60.96 | 67.43 | 73.29 | 90.54 | 72.30 |
| Railway | 84.37 | 92.39 | 74.25 | 82.83 | 74.35 | 92.72 | 86.72 | 99.53 |
| Parking lot 1 | 71.21 | 67.80 | 61.56 | 77.70 | 83.95 | 94.81 | 86.55 | 94.04 |
| Parking lot 2 | 79.53 | 76.12 | 79.32 | 80.81 | 17.45 | 15.32 | 74.74 | 92.98 |
| Tennis court | 100 | 100 | 99.53 | 99.77 | 96.27 | 99.77 | 100 | 100 |
| Running track | 98.79 | 99.09 | 98.79 | 98.64 | 90.17 | 99.70 | 98.73 | 100 |
| OA | 81.80 | 82.22 | 82.61 | 84.46 | 80.74 | 87.05 | 88.51 | **90.85** |
| AA | 84.33 | 84.54 | 84.94 | 86.51 | 80.15 | 85.58 | 89.51 | **92.50** |
| Kappa | 80.39 | 80.83 | 81.20 | 83.23 | 79.21 | 86.01 | 87.57 | **90.07** |

TABLE IV
CLASS SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TRENTO DATASET

| | SVM (H) | SVM (H+L) | ELM (H) | ELM (H+L) | Two-Branch CNN (H) | Two-Branch CNN (H+L) | Endnet (H+L) | Proposed (H+L) |
|---|---|---|---|---|---|---|---|---|
| Apple | 87.98 | 90.58 | 93.48 | 88.45 | 99.85 | 99.95 | 73.65 | 99.46 |
| Buildings | 87.87 | 95.25 | 84.64 | 97.07 | 98.56 | 99.10 | 97.12 | 94.02 |
| Ground | 97.49 | 97.49 | 94.78 | 83.51 | 93.60 | 70.13 | 90.91 | 98.93 |
| Woods | 95.33 | 99.22 | 95.13 | 99.06 | 97.45 | 99.99 | 98.27 | 99.50 |
| Vineyard | 74.32 | 87.30 | 74.86 | 82.39 | 53.76 | 97.73 | 90.67 | 99.73 |
| Roads | 63.48 | 90.36 | 64.90 | 93.57 | 63.68 | 84.44 | 90.53 | 99.31 |
| OA | 83.02 | 92.59 | 83.68 | 90.84 | 79.00 | 97.13 | 91.33 | **99.03** |
| AA | 84.41 | 93.37 | 84.63 | 90.68 | 84.48 | 91.89 | 90.19 | **98.49** |
| Kappa | 77.73 | 90.20 | 78.68 | 87.92 | 73.20 | 96.16 | 88.38 | **98.70** |

set to 300, and the learning rate is set as 0.01. As for the Endnet, the batch size is set to 32, the number of training epochs is set to 150, and the learning rate is set to 0.001. SVM is implemented using the LIBSVM toolbox. In addition, the testing and training samples are completely uniform for a fair comparison. For clarity, we define some notations before providing experimental results: H represents the HSI dataset, H + L represents the new dataset generated by the fusion of HSI and LiDAR image, ELM (H) represents running the ELM classifier for HSI data, and ELM (H + L) represents the ELM classifier is tested for the fusion dataset H + L.

Table III shows the objective results of the competing classification methods conducted on the Houston dataset in terms of the class-specific accuracy, OA, AA, and kappa coefficients. It can be observed from Table III that, for the traditional models, the ELM (H + L) achieved the best OA, AA, and kappa values that are 84.46%, 86.51%, and 83.23%, respectively. It should be noted that, for each model, the classification results obtained from the H + L dataset perform superior to those achieved from a single data source HSI. Therefore, we can draw the conclusion that the LiDAR data can offer complementary elevation information for the HSI data, and the joint use of these two data in a proper way can effectively improve the classification performance. Compared to the ELM model, the two-branch CNN model obtains inferior results on AA, which means that there are some potentials that remain to explore in the existing CNN-based methods. It is visible that the proposed MB2FscgaNet outperforms the other methods with a significant margin for the Houston data. Our proposed method obtains the optimal results in terms of OA, AA, and kappa coefficients. It can be attributed to the fact that the proposed three-stream structure can make full use of the spatial–spectral information in the HSI image and the elevation information in the LiDAR DSM image. The joint use of these two data sources can undoubtedly improve classification accuracy.

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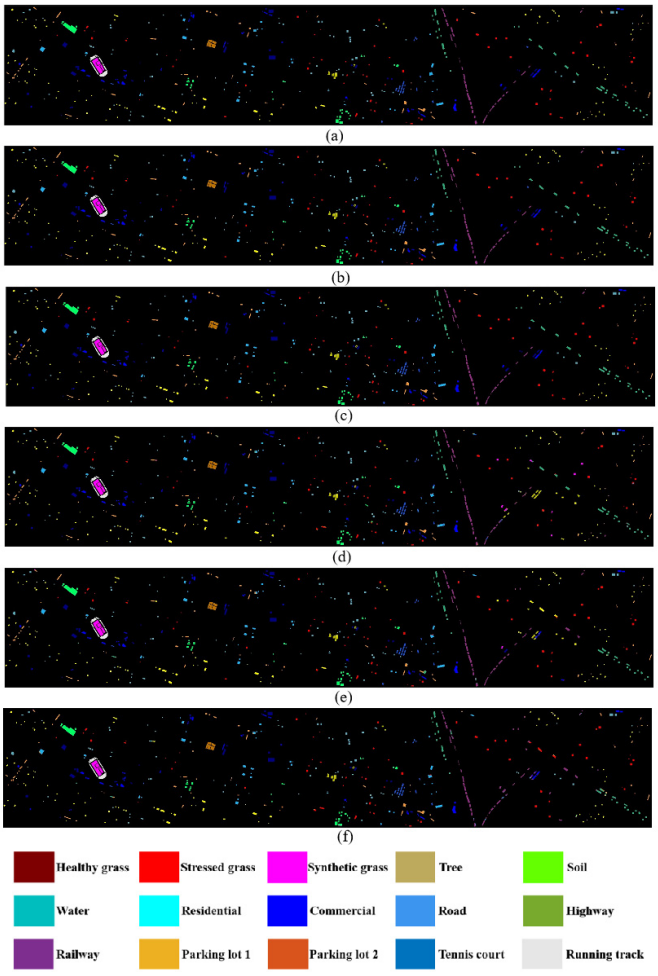



Fig. 5. Classification maps of the Houston dataset using different models. (a) Ground-truth map. (b) SVM. (c) ELM. (d) Two-branch CNN. (e) Endnet. (f) Proposed MB2FscgaNet.

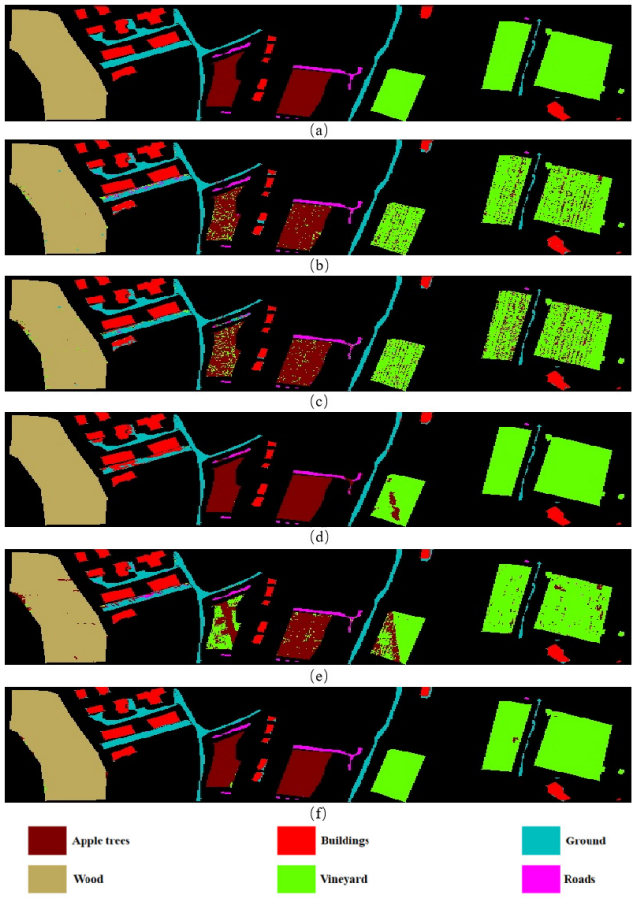



Fig. 6. Classification maps of the Trento dataset using different models. (a) Ground-truth map. (b) SVM. (c) ELM. (d) Two-branch CNN. (e) Endnet. (f) Proposed MB2FscgaNet.

For the qualitative evaluation of classification performance, Fig. 5 shows the classification maps achieved by SVM, ELM, two-branch CNN networks, EndNet, and the proposed MB2FscgaNet on the Houston dataset. The ground-truth map is also shown to facilitate the comparison. In these figures, different classes of objects are represented by different colors. The classification results of these maps are consistent with the results listed in Table III. As can be seen from Fig. 5, the SVM (H) and ELM (H) produce many outliers in the upper left part of the map. Some objects in the classification maps of the two-branch CNN network and EndNet are a little oversmoothed. The classification performance of the proposed MB2FscgaNet model is significantly better than those of the competing models.

The experimental results acquired on the Trento dataset are demonstrated in Table IV, from which it is obvious that the proposed MB2FscgaNet obtained the best classification performance. Concretely, the value of the OA gained by our newly developed MB2FscgaNet is 99.03%, which yields 1.9% and 7.7% improvements with regards to the two-branch CNN and Endnet, respectively. Compared with the standard SVM and ELM, our model also gains the improvements with 6.44% and 8.19% in OA, respectively. Such quantitative results demonstrate the effectiveness of the proposed MB2FscgaNet. Moreover, as for the ELM method, it is apparent that the OA metric using HSI is 83.68%, while the OA reaches 90.84% when utilizing H + L. This demonstrates that the use of the fused data source H + L can facilitate the model to fully exploit effective spatial and spectral-enhanced features. The excellent results of MB2FscgaNet can be attributed to the following points. On the one hand, the SCGA mechanism that is utilized in the LSHG-AB can fully integrate the complementary information of LiDAR and HSI. On the other hand, the multistream structure of the proposed MB2FscgaNet has a powerful ability for feature representation.

Similar conclusions can be drawn from Fig. 6, which presents the classification maps of the Trento dataset. From Fig. 6, we can see that the classification map that is the closest to the ground-truth map is obtained by the proposed MB2FscgaNet. Furthermore, compared with the classification maps yielded by other traditional and recently proposed methods, the proposed one exists the least mislabeled areas, especially for classes Apple trees and Vineyard. This further verifies the advantages of the proposed structure. Such classification results demonstrate that, for the traditional methods, the conversion of the LiDAR and HSI data into vectors may cause the loss of the geometric construction information.

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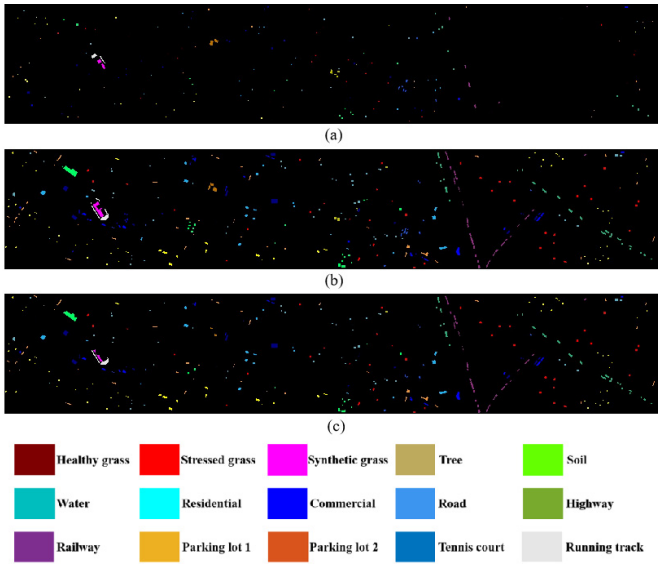



Fig. 7. Visualization of the Houston dataset. (a) Training data. (b) Testing data. (c) Testing data that are spatially separated from the training samples.

TABLE V

PERFORMANCE COMPARISON OF THE PROPOSED METHOD WITH DIFFERENT TESTING SAMPLES FOR THE HOUSTON DATASET. TEST_SSEP REPRESENTS THE TESTING SAMPLES THAT ARE SPATIALLY SEPARATED FROM THE TRAINING SAMPLES

| Method | | Test | Test_ssep |
|---|---|---|---|
| Number of Samples | | 12197 | 10364 |
| 1 | Health grass | 82.15 | 81.85 |
| 2 | Stressed grass | 83.65 | 78.83 |
| 3 | Synthetic grass | 99.80 | 100 |
| 4 | Tress | 94.41 | 93.70 |
| 5 | Soil | 100 | 100 |
| 6 | Water | 98.60 | 96.67 |
| 7 | Residual | 90.21 | 90.97 |
| 8 | Commercial | 93.54 | 93.44 |
| 9 | Road | 86.31 | 85.85 |
| 10 | Highway | 72.30 | 71.53 |
| 11 | Railway | 99.53 | 99.49 |
| 12 | Parking lot 1 | 94.04 | 93.63 |
| 13 | Parking lot 2 | 92.98 | 87.95 |
| 14 | Tennis court | 100 | 100 |
| 15 | Running track | 100 | 100 |
| | OA | 90.85 | 89.34 |
| | AA | 92.50 | 91.51 |
| | Kappa | 90.07 | 88.32 |

For other deep learning-based models, such as dual-branch CNN and Endnet, a simple cascade of features may ignore the complementary information contained in multisource data.

Since there is the spatial correlation in CNN, we compare the performance of the testing samples and the testing samples that are spatially separated from the training samples, i.e., "Test" and "Test_ssep." Figs. 7(a) and 8(a) show the training data of the Houston and Trento datasets, respectively. The testing samples of the Houston and Trento datasets are shown in Figs. 7(b) and 8(b), respectively. Since the patch size in the

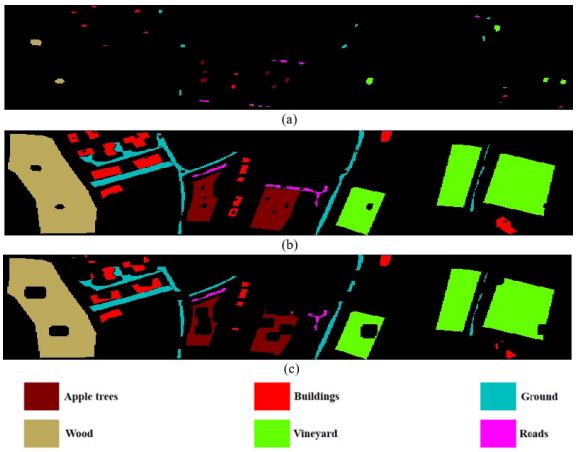



Fig. 8. Visualization of the Trento dataset. (a) Training data. (b) Testing data. (c) Testing data that are spatially separated from the training samples.

TABLE VI

PERFORMANCE COMPARISON OF THE PROPOSED METHOD WITH DIFFERENT TESTING SAMPLES FOR THE TRENTO DATASET

| Method | | Test | Test_ssep |
|---|---|---|---|
| Number of Samples | | 29595 | 25427 |
| 1 | Apple | 99.46 | 99.21 |
| 2 | Buildings | 94.02 | 93.36 |
| 3 | Ground | 98.93 | 98.32 |
| 4 | Woods | 99.50 | 99.46 |
| 5 | Vineyard | 99.73 | 99.75 |
| 6 | Roads | 99.31 | 99.23 |
| | OA | 99.03 | 99.09 |
| | AA | 98.49 | 98.22 |
| | Kappa | 98.70 | 98.73 |

experiments is $11 \times 11$, the spatially correlated receptive field of one training sample in the proposed method is $11 \times 11$. Thus, we consider the spatial correlation of the CNN and obtain the testing data that is spatially separated from the training data for the Houston and Trento datasets, as shown in Figs. 7(c) and 8(c). Tables V and VI show the performance of the proposed method with these two testing samples for the Houston and Trento datasets, respectively. As shown in Table V, the numbers of testing samples for the "Test" and "Test_ssep" on the Houston dataset are 12 197 and 10 364. The numbers of testing samples for the "Test" and "Test_ssep" on the Trento dataset are 29 595 and 25 427, respectively. From Tables V and VI, the performance of "Test_ssep" is not much worse by removing the testing data related to the training data. Especially for the Trento dataset, the OA and kappa values obtained by the "Test_ssep" are slightly better than those obtained by the "Test."

### *D. Ablation Study*

An ablation study on different configurations of the proposed model is conducted in this section. In order to study their impact on classification performance, we divide the proposed model structures into five types, i.e., Model-1 without the HSI

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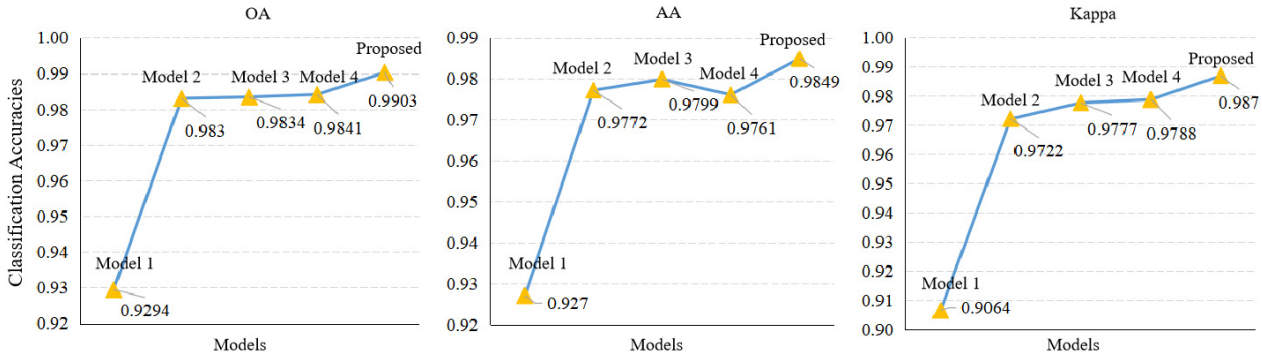



Fig. 9. Performance comparisons between the proposed model and its variants.

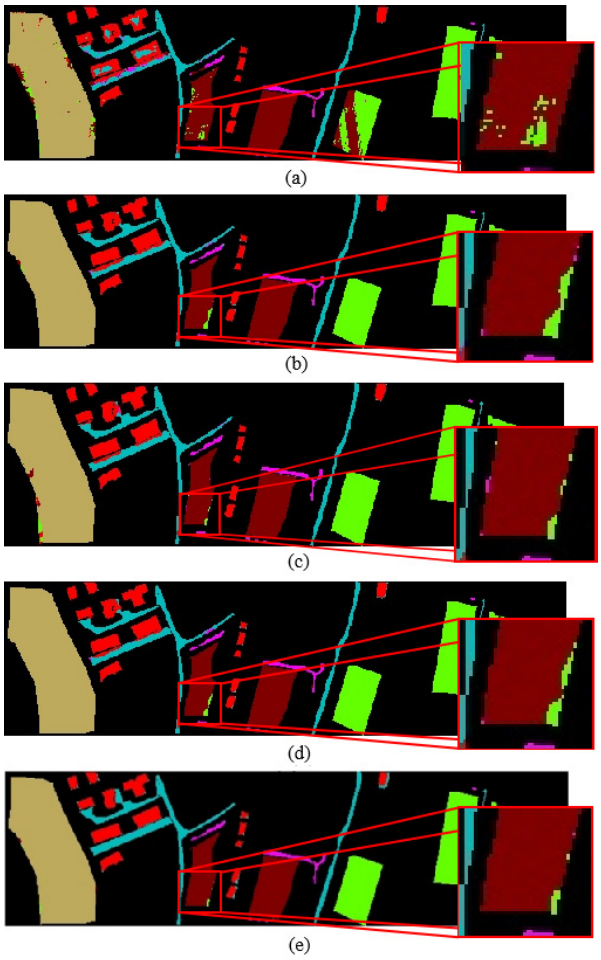



Fig. 10. Classification maps obtained by (a) Model-1, (b) Model-2, (c) Model-3, (d) Model-4, and (e) Model-5, respectively.

stream of SaSeE-FEM, Model-2 without SeSum, Model-3 that contains three main branches but no SCGA, Model-4 that contains three main branches and only self-attention, and our proposed model. We tune configuration parameters for optimal performance and adopt OA, AA, and kappa to measure their classification effect.

*1) Spatial–Spectral-Elevation Feature Extraction Module:* The HSI image not only contains rich spectral information but also spatial information, which is helpful to improve the classification accuracy. The HSI stream of SaSeE-FEM is designed to fully extract spatial information with the help of the effective LSHG-AB. As shown in Figs. 9 and 10, the comparative experimental results between Model-1 with other models prove the importance of the HSI stream in SaSeE-FEM for improving classification accuracy.

*2) Spectral Supplement Module:* We design SeSuM as one branch of MB2FscgaNet for two main reasons. First, HSI data can provide rich spectral information that can be used to distinguish materials. Second, SeSuM can further improve the distinguishability of the center pixel, which yields significant improvements in accuracy for the joint classification task. It can be seen from Figs. 9 and 10 that the use of SeSuM increases OA, AA, and kappa by 1.64%, 2.63%, and 1.77%, respectively, which shows the advantages of SeSuM.

*3) Self- and Cross-Guided Attention:* We develop SCGA for two reasons. First, the conventional convolution operation would reduce the distinguishability of the central pixel, using SCGA to help the network learn the greater weight of the interest pixels and improve the discernibility of the center pixel. Second, SCGA is used to establish mutual guidance between LiDAR and HS branches, enabling the network to produce more efficient feature transfer. To evaluate the effect of the proposed effective SCGA, we compare the proposed model (the proposed method with SCGA) with Model-3 (the proposed method without SCGA) and Model-4 (the proposed method with self-attention). As can be seen from Fig. 9, compared to Model-3 and Model-4, the proposed model obtains a better classification performance. Consistent with the conclusions in Fig. 9, the visual comparison results in Fig. 10 show that Model-3 and Model-4 generate many outliers and misclassify a lot of objects. In comparison with them, our model obtains a more homogeneous classification map, and the edges of the object are smoother. The proposed method outperforms Model 3, which proves the validity of SCGA. The proposed method also achieves better performance than Model 4, demonstrating that the innovative cross-guided attention with mutual guidance between LiDAR and HS branches is meaningful. Because human vision is not sensitive to subtle changes, we also selected a small area in the classification maps and displayed their close-up views. From the close-up views, the number of misclassified objects in the classification map obtained by our model is significantly reduced compared

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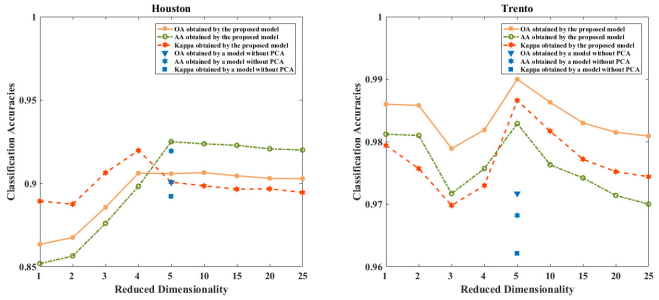



Fig. 11. Classification accuracy on different reduced dimensionality for the two datasets. (a) Houston. (b) Trento.

to Model-3 and Model-4, especially for apple trees and Vineyard.

*4) Reduced Dimensionality:* Since the HSI has redundant spectral information, we use PCA transform to reduce the spectral dimensionality of HSI in our experiments. To evaluate the effect of reduced dimensionality, we have conducted an experiment on the Houston and Trento datasets with and without PCA. We have also tested the performance of using different principal components. The results are presented in Fig. 11. As can be seen, compared with the model without PCA, the proposed model achieves higher classification accuracy. Furthermore, the proposed model achieves the best performance when the number of principal components is set to 5. Keeping increasing principal components brings only limited improvement. This is because five principal components can well represent the input image. Therefore, we decompose the input images into five principal components as the default setting.

## IV. Conclusion

In this article, we propose a novel MB2FscgaNet for the collaborative classification of multisource data (i.e., LiDAR and HSI data). For the purpose of producing improved classification performance, the proposed MB2FscgaNet adopts a multibranch CNN architecture to capture the spectral–spatial-elevation-enhanced features from HSI and LiDAR data. In the feature fusion phase, the proposed network innovatively utilizes an LSHG-AB, which provides significant spectral–spatial-elevation information guidance within and from LiDAR image to HSI, thereby yielding more effective transfer in the network. We employ an SeSuM to further supplement discriminative spectral features and enhance the discriminative ability of the center pixel. The experiments conducted on several multisource remote sensing data demonstrate that the proposed network outperforms other traditional methods, such as SVM, ELM, the two-branch CNN, and Endnet. Several ablation studies are also constructed to validate the effectiveness of the proposed architecture.

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**Wenqian Dong** (Member, IEEE) received the B.S. degree in communication engineering from Yantai University, Yantai, China, in 2014, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She was a Visiting Scholar with Simon Fraser University, Burnaby, BC, Canada, from 2018 to 2019. She is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 20 papers in refereed journals and conferences, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and the *Remote Sensing*. Her research interests include compressed sensing, machine learning, and hyperspectral remote sensing image processing.



**Tian Zhang** received the B.E. degree in communication engineering from Northwest University, Xi'an, China, in 2020. She is currently pursuing the M.S. degree with the State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an.

Her research interests include joint classification of hyperspectral images, light detection and ranging (LiDAR) images, and deep learning.



**Jiahui Qu** (Member, IEEE) received the B.S. degree in communication engineering from Yantai University, Yantai, China, in 2014, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She was an Exchange Ph.D. Student with Mississippi State University, Mississippi State, MS, USA, from 2018 to 2019, supervised by Dr. Qian Du. She is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published over 20 papers in known academic journals and conferences, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and the *Remote Sensing*. Her research interests include hyperspectral image detection, image fusion, neural networks, and deep learning.



**Song Xiao** (Member, IEEE) received the M.S. degree in communication and information system and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2001 and 2004, respectively.

From 2006 to 2007, she was a Post-Doctoral Researcher with the Viterbi School of Engineering, University of Southern California, Los Angeles, CA, USA. She is currently a Professor and the Ph.D. Director of the Beijing Electronic Science and Technology Institute, Beijing, China, and a Professor and the Ph.D. Director of Xidian University. She has authored over 80 international journal articles and conference papers. Her research interests include image compression and coding, joint source-channel coding, multimedia transmission systems over wired/wireless networks, and compressed sensing.

Dr. Xiao is also the Secretary-General of the Image Application in Military and Civilian Integration (IAMCI) Professional Committee of the China Society of Image and Graphics and a Council Member of the Shaanxi Society of Image.



**Tongzhen Zhang** received the B.E. degree in intelligence science and technology from Xidian University, Xi'an, China, in 2019, where he is currently pursuing the M.S. degree with the State Key Laboratory of Integrated Service Networks, Image Coding and Processing Center.

His research interests include hyperspectral image fusion and deep learning.



**Yunsong Li** (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He joined the School of Telecommunications Engineering, Xidian University, in 1999, where he is currently a Professor. He is also the Director of the State Key Laboratory of Integrated Service Networks, Image Coding and Processing Center, Xidian University. His research interests include image and video processing, hyperspectral image processing, and high-performance computing.

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